The headline numbers

$17.75M Total claims analyzed	**1,332** Policyholders (cleaned)	**$13.33K** Average claim	**83%** Model R^2 (variance explained)
±$3,751 Model MAE (avg. error)	**21%** Policyholders who smoke	**48%** Policyholders who are diabetic	**61%** Claims under-predicted

What we found

- Smoking status is the single biggest cost driver in the book: smokers claim roughly 4x what non-smokers claim on average **($32K vs. $8K).** This dwarfs every other factor we tested.
- BMI and age both push claims up, but far more gently than smoking - they matter for fine-tuning a price, not for flagging risk on their own.
- Diabetic status, on its own, barely moves the average claim **($13.4K vs. $13.2K).** It is not the risk signal it's often assumed to be in this dataset - it likely matters more in combination with other factors than by itself.
- Southeast carries the largest claims exposure by region **($5.8M of the $17.75M total)**, but that is driven mainly by policy volume, not by a higher average cost per policyholder.
- The data arrived in solid condition: only **8** of **1,340** records (**0.6%)** had missing values, and there were zero duplicate rows.

What the model does and where it falls short

After testing five modeling approaches (linear regression, polynomial regression, random forest, support vector regression, and XGBoost), XGBoost came out on top and was locked in as the production model. On data it has never seen, it explains **83%** of the variation in claim amounts, with a typical miss of about **$3,751**.

The one finding that matters most for Finance: the model is not neutral. It systematically leans toward under-predicting claims - roughly 6 in 10 policyholders are under-charged relative to their actual claim, and a handful of high-cost cases miss by **$15,000–$20,000**. A model that quietly under-prices risk is a slower, quieter problem than one that over-prices it, because it erodes margin without ever showing up as a customer complaint.

Where things stand against the original ask

Deliverable	Requested	Status	Notes
Cleaned dataset	Analysis-ready claims data	Complete	1,332 records, 0 duplicates, <1% missingness
Power BI dashboard	Multi-page, KPIs + segment views	Live	2 pages built; role-based access still to be configured
Predictive model	Claim cost / risk tier model	Complete	XGBoost selected; R^2 0.83, MAE $3,751

Deliverable	Requested	Status	Notes
Prediction app	User-facing input → estimate tool	Working prototype	Streamlit app; needs packaging before wider rollout

Bottom line for leadership

> The analytics foundation is solid and ready to inform pricing and reserving discussions today. The immediate priority is not more modeling - it's correcting the under-prediction bias and deciding how this tool gets governed before it's put in front of underwriters at scale. Details, evidence, and a full recommendation set follow in the sections below.